

Subquery 1 — Students scoring above the Average

```
CREATE DATABASE IF NOT EXISTS college_dbms;  
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;  
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (  
    Dept_ID INT PRIMARY KEY,  
    Dept_Name VARCHAR(50)  
);
```

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Dept_ID INT,  
    Marks INT,  
    City VARCHAR(50)  
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES  
(10, 'CSE'),  
(20, 'ECE'),  
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES  
(1, 'Vikram', 10, 85, 'Mumbai'),  
(2, 'Sneha', 20, 92, 'Bengaluru'),  
(3, 'Rohan', 10, 65, 'Mumbai'),  
(4, 'Ananya', 30, 95, 'Hyderabad'),  
(5, 'David', 20, 78, 'Bengaluru'),  
(6, 'Meera', 30, 88, 'Chennai'),  
(7, 'Arjun', 10, 72, 'Hyderabad'),  
(8, 'Mohith Gowda', 20, 90, 'Bengaluru');
```

```
SELECT Student_ID, Name, Dept_ID, Marks, City  
FROM Student
```

WHERE Marks > (SELECT AVG(Marks) FROM Student);

	Student_ID	Name	Dept_ID	Marks	City
▶	1	Vikram	10	85	Mumbai
	2	Sneha	20	92	Bengaluru
	4	Ananya	30	95	Hyderabad
	6	Meera	30	88	Chennai
	8	Mohith Gowda	20	90	Bengaluru
*	NULL	NULL	NULL	NULL	NULL

Student 4 ×

Output

Action Output

#	Time	Action
✓ 54	22:15:10	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (1, 'Vikr...

Outer:

```
CREATE DATABASE IF NOT EXISTS college_dbms;  
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;  
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (  
    Dept_ID INT PRIMARY KEY,  
    Dept_Name VARCHAR(50)  
);
```

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Dept_ID INT,  
    Marks INT,  
    City VARCHAR(50)  
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES  
(10, 'CSE'),  
(20, 'ECE'),  
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES  
(1, 'Vikram', 10, 85, 'Mumbai'),
```

```
(2, 'Sneha', 20, 92, 'Bengaluru'),
(3, 'Rohan', 10, 65, 'Mumbai'),
(4, 'Ananya', 30, 95, 'Hyderabad'),
(5, 'David', 20, 78, 'Bengaluru'),
(6, 'Meera', 30, 88, 'Chennai'),
(7, 'Arjun', 10, 72, 'Hyderabad'),
(8, 'Mohith Gowda', 20, 90, 'Bengaluru');
```

```
SELECT Name, Marks FROM Student
WHERE Marks >
(SELECT AVG(Marks) FROM Student);
```

Result Grid		
	Name	Marks
▶	Vikram	85
	Sneha	92
	Ananya	95
	Meera	88
	Mohith Gowda	90

Student 5 x

Output

Action Output

#	Time	Action
63	22:16:54	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (1, 'Vikr...

Inner:

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50)
);
```

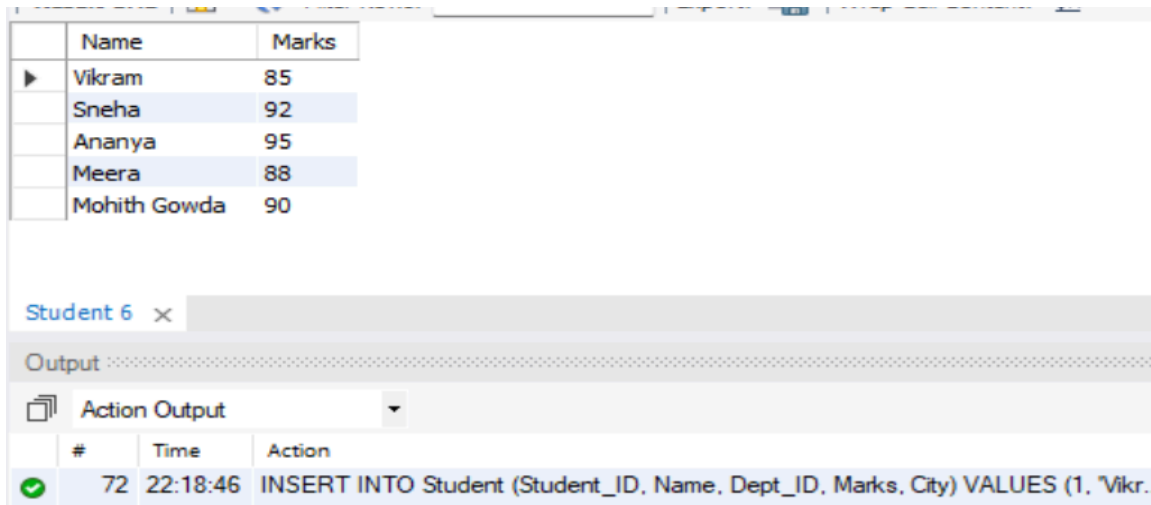
```
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Dept_ID INT,
```

```
Marks INT,  
City VARCHAR(50)  
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES  
(10, 'CSE'),  
(20, 'ECE'),  
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES  
(1, 'Vikram', 10, 85, 'Mumbai'),  
(2, 'Sneha', 20, 92, 'Bengaluru'),  
(3, 'Rohan', 10, 65, 'Mumbai'),  
(4, 'Ananya', 30, 95, 'Hyderabad'),  
(5, 'David', 20, 78, 'Bengaluru'),  
(6, 'Meera', 30, 88, 'Chennai'),  
(7, 'Arjun', 10, 72, 'Hyderabad'),  
(8, 'Mohith Gowda', 20, 90, 'Bengaluru');
```

```
SELECT Name, Marks FROM Student  
WHERE Marks >  
(SELECT AVG(Marks) FROM Student);
```



The screenshot shows a database management tool interface. At the top, a table displays the results of a query, listing students with marks greater than the average. Below this, there is a tab labeled 'Student 6'. Underneath the tab, there is an 'Output' section with a dropdown menu set to 'Action Output'. At the bottom, a log table shows the execution of an SQL statement.

	Name	Marks
▶	Vikram	85
	Sneha	92
	Ananya	95
	Meera	88
	Mohith Gowda	90

#	Time	Action
72	22:18:46	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (1, 'Vikr.

Subquery 2 — Student(s) who scored the highest marks

```
CREATE DATABASE IF NOT EXISTS college_dbms;  
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;  
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (  
    Dept_ID INT PRIMARY KEY,  
    Dept_Name VARCHAR(50)  
);
```

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Dept_ID INT,  
    Marks INT,  
    City VARCHAR(50)  
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES  
(10, 'CSE'),  
(20, 'ECE'),  
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES  
(1, 'Vikram', 10, 85, 'Mumbai'),  
(2, 'Sneha', 20, 92, 'Bengaluru'),  
(3, 'Rohan', 10, 65, 'Mumbai'),  
(4, 'Ananya', 30, 95, 'Hyderabad'),  
(5, 'David', 20, 78, 'Bengaluru'),  
(6, 'Meera', 30, 88, 'Chennai'),  
(7, 'Arjun', 10, 72, 'Hyderabad'),  
(8, 'Mohith Gowda', 20, 90, 'Bengaluru');
```

```
SELECT Student_ID, Name, Dept_ID, Marks, City  
FROM Student  
WHERE Marks = (SELECT MAX(Marks) FROM Student);
```

	Student_ID	Name	Dept_ID	Marks	City
▶	4	Ananya	30	95	Hyderabad
✱	NULL	NULL	NULL	NULL	NULL

Student 7 ✕

Output

Action Output

	#	Time	Action
✓	81	22:19:52	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (1, 'Vikr...
✓	82	22:19:52	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...

Inner:

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50)
);
```

```
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Dept_ID INT,
    Marks INT,
    City VARCHAR(50)
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
```

```
(1, 'Vikram', 10, 85, 'Mumbai'),
(2, 'Sneha', 20, 92, 'Bengaluru'),
(3, 'Rohan', 10, 65, 'Mumbai'),
(4, 'Ananya', 30, 95, 'Hyderabad'),
(5, 'David', 20, 78, 'Bengaluru'),
(6, 'Meera', 30, 88, 'Chennai'),
(7, 'Arjun', 10, 72, 'Hyderabad'),
(8, 'Mohith Gowda', 20, 90, 'Bengaluru');
```

```
SELECT MAX(Marks) FROM Student
```

Result Grid		Filter Rows:		Export:		Wrap Cell Content:
	MAX(Marks)					
	95					

Result 8 x			
Output			
Action Output			
#	Time	Action	
90	22:21:28	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES	
91	22:21:28	SELECT MAX(Marks) FROM Student LIMIT 0, 1000	

Subquery 3 — Students scoring above Rahul

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50)
);
```

```
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
```

```

Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);

```

```

INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');

```

```

INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(1, 'Vikram', 10, 85, 'Mumbai'),
(2, 'Sneha', 20, 92, 'Bengaluru'),
(3, 'Rohan', 10, 65, 'Mumbai'),
(4, 'Ananya', 30, 95, 'Hyderabad'),
(5, 'David', 20, 78, 'Bengaluru'),
(6, 'Meera', 30, 88, 'Chennai'),
(7, 'Arjun', 10, 72, 'Hyderabad'),
(8, 'Mohith Gowda', 20, 90, 'Bengaluru');

```

```

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks >
(SELECT Marks FROM Student WHERE Name = 'Rahul' OR Name = 'Mohith Gowda');

```

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
	Student_ID	Name	Dept_ID	Marks	City
▶	2	Sneha	20	92	Bengaluru
	4	Ananya	30	95	Hyderabad
✱	NULL	NULL	NULL	NULL	NULL

Student 9		×
Output		
Action Output		
#	Time	Action
✓ 99	22:23:40	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (1, 'Vikr...
✓ 100	22:23:41	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...

Subquery 4 — Students belonging to the same department as Rahul

```
CREATE DATABASE IF NOT EXISTS college_dbms;  
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;  
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (  
    Dept_ID INT PRIMARY KEY,  
    Dept_Name VARCHAR(50)  
);
```

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Dept_ID INT,  
    Marks INT,  
    City VARCHAR(50)  
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES  
(10, 'CSE'),  
(20, 'ECE'),  
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES  
(1, 'Vikram', 10, 85, 'Mumbai'),  
(2, 'Sneha', 20, 92, 'Bengaluru'),  
(3, 'Rohan', 10, 65, 'Mumbai'),  
(4, 'Ananya', 30, 95, 'Hyderabad'),  
(5, 'David', 20, 78, 'Bengaluru'),  
(6, 'Meera', 30, 88, 'Chennai'),  
(7, 'Arjun', 10, 72, 'Hyderabad'),  
(8, 'Mohith Gowda', 20, 90, 'Bengaluru');
```

```
SELECT Student_ID, Name, Dept_ID, Marks, City  
FROM Student  
WHERE Dept_ID =
```

(SELECT Dept_ID FROM Student WHERE Name = 'Mohith Gowda');

Result Grid |   Filter Rows: | Edit:    | Export/Import: 

	Student_ID	Name	Dept_ID	Marks	City
▶	2	Sneha	20	92	Bengaluru
	5	David	20	78	Bengaluru
	8	Mohith Gowda	20	90	Bengaluru
*	NULL	NULL	NULL	NULL	NULL

Student 10 ✕

Output

Action Output

	#	Time	Action
✓	108	22:25:41	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (1, "Vikr...
✓	109	22:25:41	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Dept_I...

Subquery 5 — Students who scored above the average CSE marks

```
CREATE DATABASE IF NOT EXISTS college_dbms;  
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;  
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (  
    Dept_ID INT PRIMARY KEY,  
    Dept_Name VARCHAR(50)  
);
```

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Dept_ID INT,  
    Marks INT,  
    City VARCHAR(50)  
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(1, 'Vikram', 10, 85, 'Mumbai'),
(2, 'Sneha', 20, 92, 'Bengaluru'),
(3, 'Rohan', 10, 65, 'Mumbai'),
(4, 'Ananya', 30, 95, 'Hyderabad'),
(5, 'David', 20, 78, 'Bengaluru'),
(6, 'Meera', 30, 88, 'Chennai'),
(7, 'Arjun', 10, 72, 'Hyderabad'),
(8, 'Mohith Gowda', 20, 90, 'Bengaluru');
```

```
SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks >
(SELECT AVG(Marks) FROM Student WHERE Dept_ID = 10);
```

Result Grid

Filter Rows:

Edit:

Export/Import:

	Student_ID	Name	Dept_ID	Marks	City
▶	1	Vikram	10	85	Mumbai
	2	Sneha	20	92	Bengaluru
	4	Ananya	30	95	Hyderabad
	5	David	20	78	Bengaluru
	6	Meera	30	88	Chennai
	8	Mohith Gowda	20	90	Bengaluru
*	NULL	NULL	NULL	NULL	NULL

Student 11

Output

Action Output

#	Time	Action
117	22:27:03	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (1, 'Vikr.
118	22:27:03	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks >

Subquery 6 — Students who belong to a department having a student from Mumbai

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;  
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (  
    Dept_ID INT PRIMARY KEY,  
    Dept_Name VARCHAR(50)  
);
```

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Dept_ID INT,  
    Marks INT,  
    City VARCHAR(50)  
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES  
(10, 'CSE'),  
(20, 'ECE'),  
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES  
(1, 'Vikram', 10, 85, 'Mumbai'),  
(2, 'Sneha', 20, 92, 'Bengaluru'),  
(3, 'Rohan', 10, 65, 'Mumbai'),  
(4, 'Ananya', 30, 95, 'Hyderabad'),  
(5, 'David', 20, 78, 'Bengaluru'),  
(6, 'Meera', 30, 88, 'Chennai'),  
(7, 'Arjun', 10, 72, 'Hyderabad'),  
(8, 'Mohith Gowda', 20, 90, 'Bengaluru');
```

```
SELECT Student_ID, Name, Dept_ID, Marks, City  
FROM Student  
WHERE Dept_ID IN  
(SELECT Dept_ID FROM Student WHERE City = 'Mumbai');
```

Result Grid

Filter Rows:

Edit:

Export/Import:

	Student_ID	Name	Dept_ID	Marks	City
▶	1	Vikram	10	85	Mumbai
	3	Rohan	10	65	Mumbai
	7	Arjun	10	72	Hyderabad
✱	NULL	NULL	NULL	NULL	NULL

Student 12 ✕

Output

Action Output

	#	Time	Action
✓	126	22:28:11	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (1, 'Vikr..
✓	127	22:28:11	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Dept_I...

Subquery 7 — Students who scored higher than every ECE student

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50)
);
```

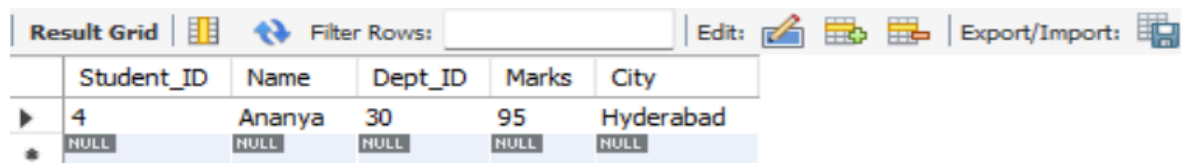
```
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Dept_ID INT,
    Marks INT,
    City VARCHAR(50)
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
```

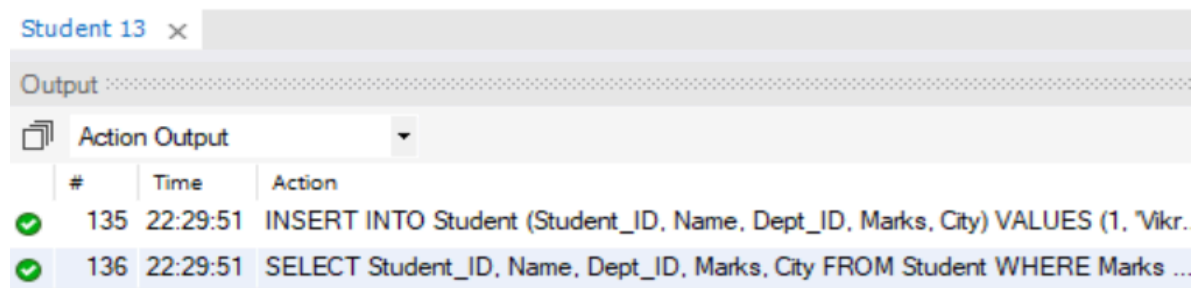
(30, 'MECH');

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(1, 'Vikram', 10, 85, 'Mumbai'),
(2, 'Sneha', 20, 92, 'Bengaluru'),
(3, 'Rohan', 10, 65, 'Mumbai'),
(4, 'Ananya', 30, 95, 'Hyderabad'),
(5, 'David', 20, 78, 'Bengaluru'),
(6, 'Meera', 30, 88, 'Chennai'),
(7, 'Arjun', 10, 72, 'Hyderabad'),
(8, 'Mohith Gowda', 20, 90, 'Bengaluru');
```

```
SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks > (
SELECT MAX(Marks)
FROM Student
WHERE Dept_ID = 20
);
```



	Student_ID	Name	Dept_ID	Marks	City
▶	4	Ananya	30	95	Hyderabad
*	NULL	NULL	NULL	NULL	NULL



#	Time	Action
✓ 135	22:29:51	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (1, 'Vikr.
✓ 136	22:29:51	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...